

Resistance and I-V characteristics (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	State the two measurements needed to find the resistance of a component.	The potential difference across it and the current through it	☐
2	Wire length and resistance: name the independent variable.	Length of the wire	☐
3	Wire length and resistance: name the dependent variable.	Resistance of the wire	☐
4 ★	Wire length and resistance: name three control variables.	The temperature, thickness and material of the wire	☐
5 ★	Describe how the length of a wire affects its resistance.	The resistance increases (it is directly proportional to length).	☐
6 ★	In the wire-length practical, the fixed crocodile clip is at the 1 cm mark, but every length is read from 0 cm. Name the type of error this causes.	A zero error (a systematic error): every length is 1 cm too long.	☐
7 ★	Explain why the potential difference is kept small when measuring the resistance of a wire.	So the current is small and the wire does not heat up.	☐
8	What does it mean for two resistors to be in series?	They are connected one after the other, on a single path.	☐
9	What does it mean for two resistors to be in parallel?	They are on separate branches.	☐
10 ★	Describe how to calculate the total resistance of two resistors in series.	Add the two resistances: $R_{\text{total}} = R_1 + R_2$.	☐

QUESTION		ANSWER	✓
11	A lamp, a power supply and a variable resistor are in series. State two ways to increase the potential difference across the lamp.	Increase the supply potential difference. Decrease the resistance of the variable resistor.	☐
12 ★	Finding the I–V characteristic of a component: name the independent and dependent variables.	Independent: potential difference across the component Dependent: current through it	☐
13	Describe the graph of resistance against length for a wire.	A straight line through the origin.	☐